
Neural Approximate Mirror Maps for DC Optimal Power Flow

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Abstract

Optimal Power Flow (OPF) is a fundamental optimization problem in power systems, aiming to determine generator dispatch decisions that minimize operational cost while satisfying physical and operational constraints. Recent diffusion-based approaches have demonstrated the ability to model the distribution of dispatch solutions and capture the multi-modal structure of feasible operating points. However, these methods often struggle to consistently satisfy physical constraints during generation, requiring projection or post-processing methods to ensure feasibility. In this work, we explore the use of Neural Approximate Mirror Maps (NAMMs) as a framework for improving feasibility in diffusion-based DC-OPF. NAMMs learn an approximately invertible transformation between the constrained OPF feasible set and an unconstrained latent mirror space, allowing diffusion models to work in a representation that better reflects the underlying geometry of feasible solutions. We integrate the learned mirror maps into a diffusion-based OPF pipeline and evaluate their impact on constraint satisfaction and sample generation. Our experiments demonstrate that diffusion performed in the learned mirror space produces samples with improved feasibility and reduced constraint violations compared to diffusion models trained directly in the original dispatch space. These results suggest that a learned optimization geometry can help incorporate power-system constraints into generative models. Future work will extend this framework to the significantly more challenging AC-OPF setting, where nonlinear and non-convex power flow equations further complicate the structure of the feasible region.

1 Introduction

Optimal Power Flow (OPF) is a critical optimization problem in electric power systems that determines generator dispatch decisions while satisfying network physics and operational constraints. As modern power grids become increasingly complex due to factors such as renewable integration, distributed energy resources, and uncertain demand patterns, solving OPF efficiently and reliably continues to be an important challenge [2]. While the DC-OPF formulation is significantly more tractable than its AC counterpart, its feasible region is still defined by a collection of operational constraints that must be respected to ensure physically meaningful solutions.

Recent advances in machine learning have motivated a shift from solving optimization problems independently to learning representations of their solution spaces directly from data. In particular, generative approaches have emerged as a promising framework for modeling distributions over feasible operating points. However, many generative models operate in Euclidean space and do not naturally account for the geometry caused by the problem constraints. As a result, produced samples often violate feasibility requirements.

A promising direction for addressing this challenge is to incorporate the geometry of the feasible region directly into the learning process. Mirror descent provides a geometry-aware alternative

through the use of mirror maps, which directly encode the structure of the underlying constraint set [6]. Unfortunately, deriving suitable mirror maps analytically is difficult for complex constrained systems like OPF. This generates the need for alternative solutions.

Neural Approximate Mirror Maps (NAMMs) provide a possible approach by learning an approximately invertible transformation between a constrained feasible set and an unconstrained latent space [3]. By leveraging Input Convex Neural Networks (ICNNs) and cycle-consistent inverse mappings, NAMMs learn representations that capture the geometry of complex constrained solution spaces while enabling efficient optimization and sample generation [1].

In this project, we investigate the application of NAMMs to DC-OPF. We compare representations in the original dispatch space and the learned mirror space, evaluating their ability to preserve feasibility and capture the underlying structure of OPF solutions. Our results suggest that mirror-space representations better reflect the geometry of the feasible set and produce samples with improved constraint satisfaction compared to methods operating directly in the original solution space. These findings highlight the potential of learned mirror geometry for other constrained optimization problems and motivate extensions to the more challenging AC-OPF setting.

2 Related Work

2.1 Optimal Power Flow

Optimal Power Flow (OPF) is one of the most widely studied optimization problems in power systems. Given network topology and system demand, OPF determines generator dispatch decisions that minimize an objective, typically generation cost, while satisfying physical power flow equations and operational constraints. The AC-OPF formulation is inherently nonlinear and non-convex, making large-scale optimization computationally challenging. Consequently, many practical studies employ the DC approximation, which linearizes the power flow equations and results in a convex optimization problem that can be solved efficiently using standard optimization techniques.

Classical OPF solvers rely on mathematical programming approaches including interior-point and quadratic programming methods [2]. These methods provide strong optimality guarantees and remain the industry standard for power system operation. However, solving a new optimization problem for every operating condition can be computationally expensive in large-scale settings and motivates the development of learned representations of OPF solution spaces.

2.2 Learning-Based Approaches for OPF

Recent work has explored machine learning approaches for approximating OPF solutions directly from data. Neural networks have been used to predict dispatch decisions, estimate optimal costs, and accelerate optimization procedures [5]. More recently, generative modeling approaches have been proposed to capture the distribution of feasible operating points rather than producing a single deterministic solution.

Among these approaches, DiffOPF formulates OPF as a conditional generative modeling problem using diffusion models [4]. Instead of learning a single mapping from load conditions to dispatch decisions, DiffOPF learns a distribution over feasible solutions and can generate multiple warm start operating points for the same system conditions. This approach is attractive because practical OPF instances often have multiple feasible dispatch configurations with similar costs. However, diffusion models operating in the original solution space generally do not enforce physical constraints by construction, potentially producing infeasible samples that require additional correction procedures. Although our final implementation does not incorporate the DiffOPF pipeline, its formulation motivates the broader question of how constraint geometry can be incorporated into generative models for power systems.

2.3 Input Convex Neural Networks

Input Convex Neural Networks (ICNNs) provide a neural architecture whose output is convex with respect to its input [1]. Originally introduced as a framework for learning convex functions, ICNNs have found applications in optimization and geometry-aware learning. Their convexity properties

make them applicable for mirror-descent-based methods, where convex potentials induce mappings between primal and dual spaces.

A practical challenge when using ICNNs as mirror potentials is that convexity alone does not guarantee strict convexity, which is often required to obtain stable and invertible mirror mappings. A common approach is to augment the learned potential with a quadratic regularization term of the form

$$\frac{\lambda}{2} \|x\|_2^2.$$

This addition enforces strong convexity, ensuring that the gradient map is injective and improving numerical stability during optimization and inversion [6].

By combining the representational flexibility of ICNNs with strong-convexity regularization, learned mirror maps can capture the geometry of complex feasible sets while retaining many of the theoretical properties underlying mirror descent. These characteristics make ICNNs a natural foundation for geometry-aware generative modeling in constrained optimization problems such as OPF.

2.4 Neural Approximate Mirror Maps

Mirror descent is a geometry-aware optimization method that generalizes gradient descent by replacing Euclidean distance with a Bregman divergence induced by a mirror map. The effectiveness of mirror descent depends critically on the choice of mirror map, which must typically be derived analytically for a specific constraint set. Recent work has explored parameterizing mirror maps using Input Convex Neural Networks (ICNNs), which provide a flexible framework for learning convex potentials directly from data [6]. However, learning mirror maps that accurately capture the geometry of complex constrained domains continues to be a challenging problem.

Neural Approximate Mirror Maps (NAMMs) address this limitation by learning mirror maps directly from data [3]. Rather than specifying an analytic mirror potential, NAMMs learn a forward map from a constrained manifold to an unconstrained latent space and an approximate inverse map back to the original domain. Training combines cycle-consistency objectives, feasibility constraints, and regularization terms to ensure that the learned transformations remain stable and approximately invertible. By learning geometry directly from samples, NAMMs provide a flexible framework for representing nonlinear feasible sets without requiring explicit mathematical characterization.

While NAMMs were originally proposed as a preprocessing framework for constrained diffusion models, their ability to learn feasible-set geometry makes them independently interesting for power system applications. Our work explores this setting by applying NAMMs directly to DC-OPF feasible solutions and evaluating the quality of the resulting mirror-space representation.

3 Research Questions and Approach

3.1 Research Questions

The primary objective of this project is not to solve DC-OPF more efficiently than existing optimization algorithms. For the IEEE 118-bus system considered in this work, standard DC-OPF solvers are already capable of computing optimal solutions rapidly while guaranteeing feasibility and optimality. Instead, this project investigates a more fundamental machine learning question, which is whether or not a learned representation of the OPF feasible region can improve generative modeling of power-system operating points. This motivates three research questions:

1. Can a NAMM learn a meaningful representation of the DC-OPF feasible region?
2. Does diffusion in the learned mirror space produce more feasible samples than diffusion performed directly in OPF sample space?
3. What tradeoffs emerge between feasibility and reconstruction quality?

The first question relates to representation learning. Before a mirror-space diffusion model can succeed, the learned forward and inverse maps must approximately preserve OPF solutions and map latent samples back toward the feasible set. The second question evaluates the practical usefulness of the representation. If the mirror map can capture meaningful constraint geometry, then a diffusion

model trained in mirror space should generate samples with fewer physical violations than an equivalent model trained directly on the original OPF vectors. The third question is motivated by the observation that feasibility and distribution matching are not necessarily aligned objectives. A model may reproduce global statistics of the dataset, but still produce largely infeasible operating points.

3.2 Problem Formulation

3.2.1 Data Generation

The experiments are performed on the IEEE 118-bus test system using a DC-OPF formulation. Training data are generated by repeatedly perturbing load values and generator cost coefficients before solving a new DC-OPF instance. Each solution is recorded as a vector

$$x = [P_{d,1}, \dots, P_{d,99}, P_{g,1}, \dots, P_{g,54}] \in \mathbb{R}^{153}$$

where the first 99 dimensions correspond to active load values and the final 54 dimensions correspond to generator dispatch values.

The resulting dataset can be viewed as an empirical sample from the feasible DC-OPF solution set. Every training point is produced by a classical optimization solver and therefore satisfies the DC-OPF constraints used during data generation. The objective of the learned model is not to rediscover the optimization procedure itself, but rather to learn the structure of this feasible solution set.

Feasibility is evaluated through a soft constraint function composed of four components:

$$\mathcal{L}_{\text{bound}} + \mathcal{L}_{\text{balance}} + \mathcal{L}_{\text{load}} + \mathcal{L}_{\text{line}}$$

The generator-bound term penalizes dispatch values that exceed generator operating limits, while the power-balance term penalizes mismatch between total generation and total load. Additional penalties discourage negative load values and transmission line-flow violations computed using the PTDF matrix exported during data generation. These losses are not used as exact projections. Instead, they provide a continuous measure of physical infeasibility that can be incorporated into NAMM training and evaluation.

3.2.2 Neural Approximate Mirror Maps

The first stage of the pipeline learns a pair of transformations between OPF space and a latent mirror space. The forward map

$$g_\phi : \mathcal{X} \rightarrow \mathcal{Y}$$

transforms OPF vectors into mirror-space coordinates, while the inverse map

$$f_\psi : \mathcal{Y} \rightarrow \mathcal{X}$$

maps latent samples back into OPF space.

Following the original NAMM framework, the forward map is implemented using an ICNN [3]. Rather than learning an arbitrary transformation, the forward map is parameterized through a convex potential whose gradient defines the mirror representation. This design choice reflects the connection between mirror descent and convex geometry. It allows us to have a structured representation that is more stable than an unconstrained neural mapping.

The inverse map is implemented as a residual multilayer perceptron. Unlike the forward map, the inverse transformation does not need to preserve convexity and thus has greater flexibility. The goal of the inverse network is to reconstruct feasible OPF vectors from mirror-space representations while correcting deviations that arise from noise and sampling.

For training, we optimize a weighted combination of three objectives:

$$\mathcal{L}_{\text{cycle}}(g_\phi, f_\psi) + \lambda_{\text{constr}} \mathcal{L}_{\text{constr}}(g_\phi, f_\psi) + \lambda_{\text{reg}} \mathcal{R}(g_\phi)$$

The cycle-consistency loss $\mathcal{L}_{\text{cycle}}$ enforces approximate invertibility, ensuring that information is preserved in both directions. The constraint loss $\mathcal{L}_{\text{constr}}$ evaluates feasibility after inverse mapping. During training, noisy mirror-space samples are passed through the inverse network and penalized

according to the physical constraint function. This encourages the inverse map to act as a feasibility-restoring transformation rather than merely a decoder. Finally, the regularization term $\mathcal{R}(g_\phi)$ prevents the learned representation from becoming unnecessarily distorted and stabilizes optimization. The resulting mirror space can be interpreted as a learned coordinate system whose geometry is adapted to the OPF feasible set.

3.2.3 Mirror Diffusion Model

Once the NAMM has been trained, its parameters are frozen and used to construct a mirror-space dataset. Every OPF vector is transformed according to

$$y = g_\phi(x),$$

and the resulting mirror-space samples become the training data for the diffusion model. The diffusion implementation follows a variance-preserving stochastic differential equation (VP-SDE) formulation. During training, Gaussian noise is added to mirror-space samples at random timesteps, and a score network is trained to predict the corresponding denoising direction. The score model itself is implemented as a time-conditioned multilayer perceptron.

Conceptually, the diffusion model learns the score

$$\nabla_y \log p_t(y),$$

which describes how samples should move during reverse diffusion. Starting from Gaussian noise, repeated denoising steps gradually transform the sample into a structured mirror-space point.

After generation, the inverse mirror map converts the sample back into OPF space,

$$f_\psi(\hat{y}).$$

3.3 Conditional Generation and Baseline Comparison

The most practically relevant evaluation setting is conditional generation. In real power-system operation, load values are observed inputs while generator dispatch decisions must be determined in response. As such, the conditional task fixes the load coordinates and generates the corresponding dispatch values.

Conditioning is implemented using an inpainting-style procedure during reverse diffusion. After each denoising step, the current sample is mapped back into OPF space, the load coordinates are replaced with the true observed loads, and the modified vector is mapped forward again into mirror space. This process ensures that conditioning information remains present throughout sampling while still allowing the diffusion model to generate dispatch values.

To evaluate whether the mirror representation provides value beyond diffusion alone, we compare against a sample-space diffusion baseline. The baseline uses the same VP-SDE framework and a similar score network but is trained directly on OPF vectors rather than mirror-space representations. This comparison isolates the effect of the learned geometry.

4 Experiments and Results

We compare diffusion performed directly in the original dispatch space against diffusion performed in a learned NAMM mirror space. Generated samples are evaluated using the feasibility loss defined in Section 3 together with generator-dispatch metrics computed on held-out test data.

4.1 Quantitative Results

Table 1: Quantitative comparison between diffusion performed directly in dispatch space and diffusion performed in a learned NAMM mirror space.

Metric	Sample Space	NAMM
Mean Constraint Loss	36.50	2.82
Max Constraint Loss	1035.63	5.52
Mean Bound Loss	27.76	2.55
Mean Balance Loss	8.59	0.27
Mean P_g MAE (MW)	116.3	70.2
Median P_g MAE (MW)	91.9	66.0
Samples with MAE < 50 MW	6.6%	23.8%
Samples with MAE < 100 MW	57.4%	86.7%

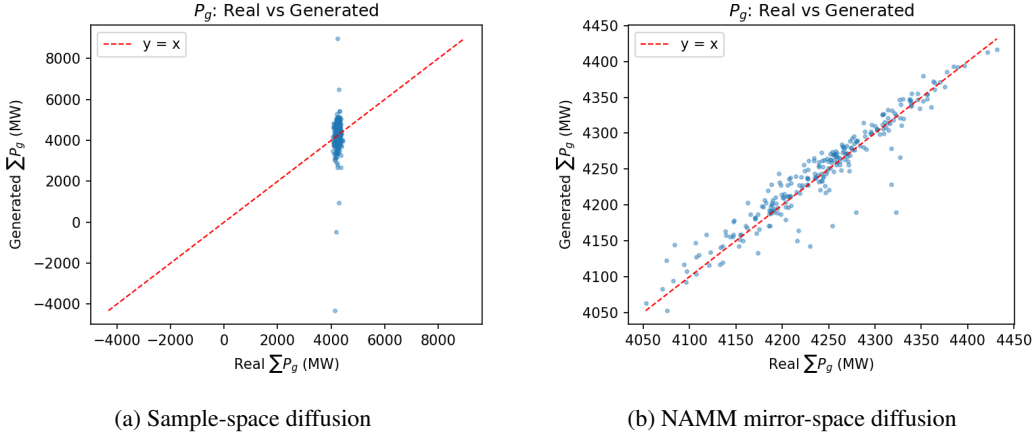


Figure 1: Comparison of generated and real total generation values.

Preservation of Total Generation. Figure 1 compares generated and real total generation values. While the sample-space diffusion model exhibits significant dispersion and several extreme outliers, the NAMM model remains tightly concentrated around the identity line. This indicates that diffusion in the learned mirror space more accurately preserves the structure of the underlying OPF solution distribution.

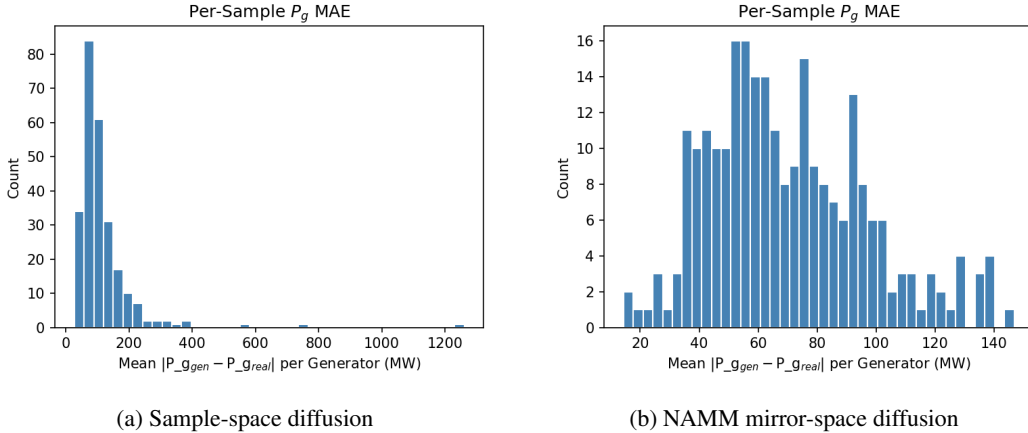


Figure 2: Distribution of per-sample generator-dispatch mean absolute error (MAE).

Generator Dispatch Accuracy. Figure 2 compares per-sample generator-dispatch errors between the two approaches. The NAMM model shifts the error distribution toward lower values and substantially reduces the frequency of large generator-dispatch deviations. As shown in Table 1, the average generator-dispatch MAE decreases from 116.3 MW for the sample-space baseline to 70.2 MW for the NAMM model, representing an improvement of approximately 40%.

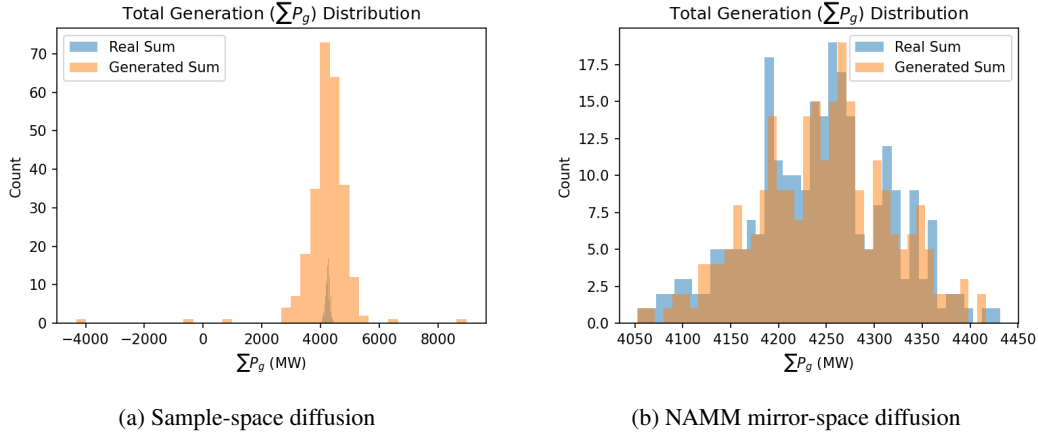


Figure 3: Distribution of total generations.

Distribution of Total Generation. Figure 3 compares the distributions of total generated power across the test set and generated samples. The sample-space diffusion model produces a distribution with substantially larger variance and several extreme outliers, leading to a much wider spread than observed in the real data. In contrast, diffusion in the NAMM mirror space generates a distribution whose shape closely matches the real distribution, with similar center, spread, and overall support. The strong overlap between the generated and real histograms suggests that the learned mirror representation better captures the underlying structure of feasible OPF solutions while avoiding the unrealistic extreme values produced by the sample-space baseline.

Overall, the results demonstrate that performing diffusion in a learned mirror-space representation improves both feasibility and dispatch accuracy. The largest gains are observed in generator-bound and power-balance constraints, suggesting that the learned geometry provided by NAMMs more effectively captures the structure of the DC-OPF feasible region than diffusion performed directly in the original dispatch space.

5 Discussion and Conclusions

5.1 Results and Implications

The experimental results demonstrate that incorporating NAMMs into the diffusion pipeline substantially improves both feasibility and generator-dispatch accuracy. Across all major metrics, diffusion performed in the learned mirror space consistently outperforms diffusion performed directly in the original dispatch coordinates.

Note that the largest improvements are observed in the generator-bound and power-balance constraints, which constitute the dominant sources of infeasibility in the sample-space baseline. The average constraint loss decreases by more than an order of magnitude, while extreme infeasible outliers are largely eliminated. At the same time, the NAMM model achieves lower generator-dispatch error and produces samples whose aggregate generation statistics more closely match those of the real OPF solutions.

These findings suggest that the benefits of NAMMs may extend beyond simple representation learning. By transforming the constrained OPF feasible region into a geometry-aware latent space, the mirror map provides a coordinate system that is more compatible with the diffusion process. As a result, the generative model is able to better preserve the structure of the feasible set while maintaining sample

quality. More broadly, these results indicate that incorporating optimization geometry into generative modeling may be a promising direction for constrained scientific machine learning problems.

5.2 Limitations

Despite the observed improvements, several limitations remain. First, the generated samples are not perfectly feasible. Although NAMMs significantly reduce constraint violations, the resulting samples still exhibit nonzero generator-bound and power-balance errors. As such, the learned mirror map should be viewed as an approximation to the feasible geometry rather than an exact feasibility-preserving transformation.

Second, the experiments are restricted to the DC-OPF setting. DC-OPF provides a simplified linear approximation of power-system behavior and omits many of the nonlinear phenomena present in real transmission networks. As a result, improvements observed in this work may not directly transfer to more realistic AC-OPF formulations.

Finally, the evaluation is conducted on a single benchmark system. Additional experiments on larger networks and alternative operating conditions would be necessary to fully characterize the scalability and robustness of the proposed approach.

5.3 Future Work

Several directions remain for future investigation. The most immediate extension is the application of NAMM-based diffusion to AC-OPF. Compared to DC-OPF, AC-OPF introduces nonlinear power-flow equations, voltage constraints, and a highly non-convex feasible region. These characteristics make AC-OPF a substantially more challenging generative modeling problem and provide a natural setting in which geometry-aware representations may offer even greater benefits. Future work could also explore alternative mirror-map architectures and regularization strategies that may improve invertibility and further reduce residual constraint violations.

Beyond power systems, the proposed framework may be applicable to a broader class of constrained optimization problems in which feasible solutions occupy a complex low-dimensional region embedded within a higher-dimensional space.

5.4 Conclusion

In conclusion, this project investigated the use of NAMMs as a geometry-aware representation for diffusion-based DC-OPF generation. By learning an approximately invertible transformation between the constrained OPF feasible region and an unconstrained mirror space, NAMMs enable diffusion models to operate in a representation that better reflects the underlying structure of feasible solutions.

Experimental results on the IEEE 118-bus system demonstrate that mirror-space diffusion substantially improves feasibility while reducing generator-dispatch error compared to diffusion performed directly in the original dispatch space. These findings suggest that the learned optimization geometry can improve the ability of generative models to respect physical constraints. They also motivate further investigation of mirror-space generative modeling for increasingly complex power-system optimization problems.

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Appendix:

A Data Analysis

All experiments are conducted on a synthetic DC-OPF dataset generated from the IEEE 118-bus benchmark network. The system contains 118 buses, 99 active loads, and 54 generators. For each sample, load demands and generator cost coefficients are randomly perturbed before solving a new OPF problem. The resulting dataset consists of feasible operating points represented by the concatenated load and generator dispatch vector

$$x = [P_{d,1}, \dots, P_{d,99}, P_{g,1}, \dots, P_{g,54}] \in \mathbb{R}^{153}$$

Because every sample is obtained from a solved OPF instance, all datapoints satisfy the underlying power balance and generator operating constraints. The dataset therefore represents samples drawn directly from the feasible OPF solution manifold rather than arbitrary points in the dispatch space.

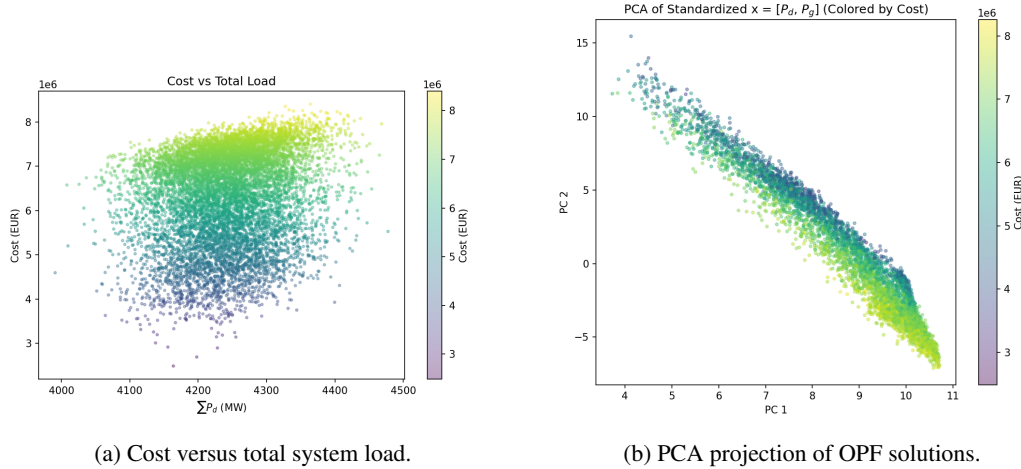


Figure 4: Global statistics of the IEEE 118-bus OPF dataset.

Figure 4 summarizes several global properties of the dataset. Figure 4a shows the relationship between total system demand and operating cost. As expected, higher aggregate demand generally results in higher operating costs. However, the substantial vertical spread suggests that cost is also strongly influenced by generator availability, dispatch decisions, and network constraints. Figure 4b presents a two-dimensional PCA projection of the standardized OPF vectors. The samples lie on a highly structured curved manifold rather than filling the ambient 153-dimensional space uniformly, indicating strong correlations and nonlinear dependencies between loads and generator dispatch levels.

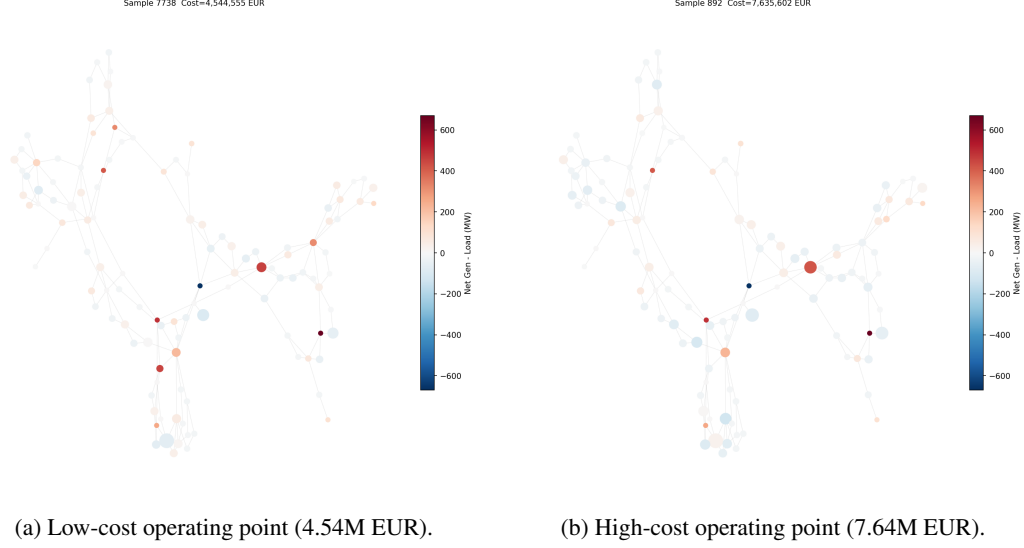


Figure 5: Representative OPF solutions from the IEEE 118-bus dataset.

Figure 5 visualizes two representative OPF solutions from the dataset. Node color indicates net generation minus load, while node size reflects the magnitude of power injection or withdrawal. The low-cost operating point in Figure 5a exhibits a dispatch pattern that efficiently satisfies demand using comparatively inexpensive generation resources. In contrast, the higher-cost solution in Figure 5b requires a substantially different allocation of generation across the network, illustrating the diversity of feasible operating conditions present in the dataset. These examples highlight that multiple physically feasible dispatch configurations can exist for similar overall demand levels, motivating the need for generative models that capture the full distribution of OPF solutions rather than a single deterministic operating point.

B Unconditional Sampling

Unlike the experiments in the main paper, which compare conditional generation quality against held-out OPF solutions, the unconditional setting requires the model to generate complete operating points without conditioning on a prescribed load profile. Consequently, the model must learn the joint distribution of both load demands and generator dispatch values directly from data.

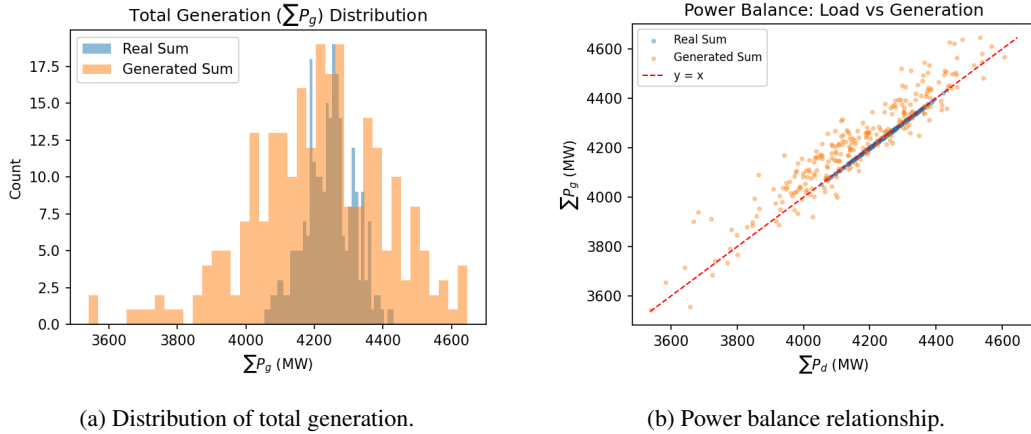


Figure 6: Unconditional generation results.

Figure 6a compares the distributions of total generation and power balance statistics between real and generated samples. Because the model is free to sample from the full learned distribution,

the generated operating points exhibit a wider spread than the held-out test set. Nevertheless, the generated samples remain concentrated in the same operating region as the real data and preserve the overall structure of the underlying distribution.

In particular, Figure 6b demonstrates that generated samples continue to satisfy the fundamental relationship between total load and total generation. While the generated points exhibit greater variability than the test data, they remain closely aligned with the identity line, indicating that the model has learned the power-balance structure present in the dataset.

C Ablation Study: Normalized vs. Unnormalized Data

To evaluate the impact of feature normalization, we train the same NAMM diffusion pipeline both with and without normalization of the OPF variables. Normalization is commonly used when training generative models on heterogeneous datasets because different variables can have substantially different numerical scales. In the IEEE-118 DC-OPF dataset, generator outputs, load demands, and auxiliary quantities span different ranges, which may influence optimization dynamics during training.

Table 2 summarizes the quantitative results. The model trained without normalization achieves a lower MMD score, suggesting slightly improved distributional matching in the original feature space. However, this improvement comes at the expense of substantially worse physical feasibility. The mean constraint loss increases from 3.96 MW to 15.71 MW, while the maximum observed violation grows from 10.38 MW to 32.76 MW. Most of this degradation comes from generator-bound violations, where the mean value increases by more than a factor of six.

Table 2: Normalization ablation study comparing training in the normalized feature space against training directly in the original OPF variable space.

Metric	Normalized	No Normalization
Mean Constraint Loss	3.96	15.71
Max Constraint Loss	10.38	32.76
Mean Bound Loss	2.50	15.40
Mean Balance Loss	1.30	0.31
Mean Load Loss	0.16	0.00
MMD (RBF)	0.49	0.23

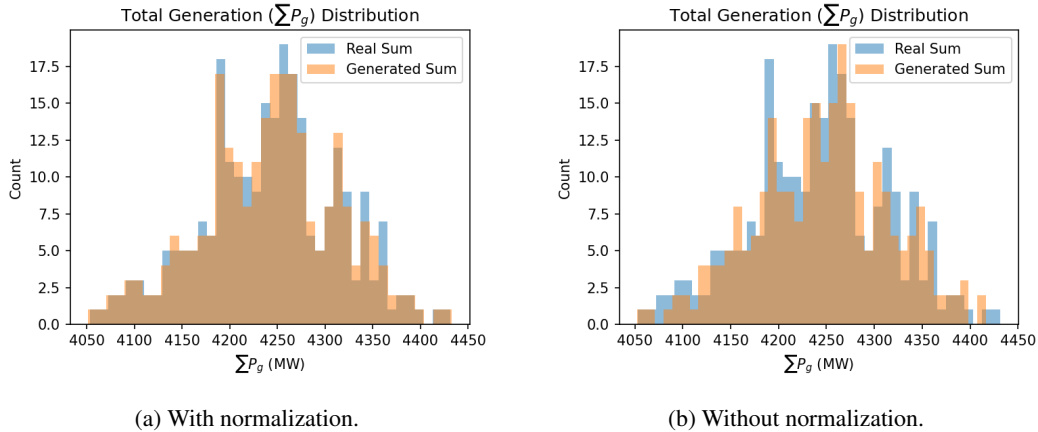


Figure 7: Distribution of total generated power ($\sum P_g$) under the normalization ablation.

Figure 7 compares the distributions of total generated power. Both models successfully capture the overall operating range of the OPF solutions and produce samples centered around the true distribution. However, the unnormalized model exhibits a noticeably broader spread, generating a

larger number of samples in low-density regions of the state space. This behavior is consistent with the lower MMD score reported in Table 2, but does not necessarily imply improved physical realism.

Overall, these results demonstrate that normalization is an important component of the proposed framework. Although training without normalization can improve certain distribution-matching metrics, it significantly degrades physical feasibility. The normalized representation therefore provides a more favorable balance between realistic sample generation and satisfaction of the underlying OPF constraints.

D Code Availability

The code used for data generation, NAMM training, diffusion modeling, and evaluation is publicly available at:

<https://github.com/ryanhu00/namm-opf>

The repository includes scripts for dataset generation, model training, conditional and unconditional sampling, evaluation, and figure reproduction.